

ORIGINAL ARTICLE

ECONOMICS OF TRANSITION
AND INSTITUTIONAL CHANGE

WILEY

Do bigger farms suffer less from corruption? Anti-corruption efforts and the recovery of livestock production

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Email: rrakhmetov.at2@gmail.com**Abstract**

A transparent and reliable business environment, free from corruption, is essential for agricultural development. However, the impact of corruption control on productive activities remains ambiguous. While corruption can facilitate transactions in an overly regulated environment, its control is also crucial for creating transparency and fostering economic progress. This paper examines how different organizational farm structures respond to corruption control in a long-term dynamic specification. Using a unique dataset of an objective corruption control measure and livestock production at the subnational level in Kazakhstan and Russia, the analysis reveals that the relationship between production growth and corruption control is non-linear and differs across household farms, peasant farmers and agricultural enterprises. We find peasant farms exhibit superior average growth in regions with stronger corruption control than other farm structures in regions with weaker corruption control. Conversely, enterprises report lower production growth in regions with intensified anti-corruption measures in the short run, yet they eventually benefit from stronger corruption control in the long run. Livestock production by

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small household producers positively correlates with anti-corruption initiatives, but this association wanes over time. This paper unravels the intricate dynamics surrounding corruption control and different business structures, shedding light on the multifaceted nature of their interrelationship.

KEYWORDS

corruption control, livestock production, organizational structures, production growth

JEL CLASSIFICATION

D73, Q18, P32, O13

1 | INTRODUCTION

The influence of corruption on aggregated production and economic growth has received great attention among scholars during the past decade (e.g. Aidt, 2009; Gründler & Potrafke, 2019). However, the relationship between corruption and productive activities is theoretically and empirically ambiguous. On the one hand, corruption impedes economic operations (Asiedu & Freeman, 2009; Buchanan & Tullock, 1962; Rose-Ackerman & Truex, 2012); on the other hand, corruption occasionally can help progress in overly bureaucratic environments (Egger & Winner, 2005; Leff, 1964; Levy, 2007). If the evidence of corruption's inconsistency with productive activities does not imply that corruption is always detrimental, then what organizational form of businesses gains the most from more intensive corruption control?

Corruption in the agricultural sector has been investigated in numerous aspects, including the effort to associate rent-seeking with the misallocation of subsidies (Kvartiuk & Herzfeld, 2021; Tambulasi, 2009; Teichmann et al., 2020), an attempt to link bribery and inefficient land use (Barbier, 2012; Bulte et al., 2007), and relate the corruption to the unsustainable water management in the irrigation sector (Jacoby et al., 2021; Suhardiman & Mollinga, 2017). Previous research has highlighted that in the agri-food sector, corruption is perceived as a major impediment to business operations, much like in other industries (Herzfeld et al., 2018).

Recent analyses beyond national-level corruption indicators present contradictory findings regarding the impact of corruption control on economic progress. The analyses focusing exclusively on China provide evidence that more vigorous anti-corruption efforts, measured by registered cases and convictions, encourage foreign direct investments (Cole et al., 2009), increase the likelihood of firms making long-term investments (Xu & Yano, 2017) and improve firms productivity (Kong et al., 2020). In contrast, other scholars find evidence for Russia that regions with more registered crime cases of bribe-taking and incidents of experienced corruption attract fewer investments (Zakharov, 2019). Johnson et al. (2011) demonstrate an adverse effect of corruption, measured by convictions, on state-level gross domestic product per worker growth rates for the USA.

The contrasting results for productive activities within different corruption control environments might be attributable to the fact that the utilized data covers all producers, regardless

of their organizational form. All studies implicitly assume that firms are equally exposed to potentially corrupt officials or have the same financial means to address requests for personal favours. Whereas data limitations prevent being more specific, this suggests significant non-linearities in the relation between productive activities and anti-corruption efforts. In conjunction, when cross-industry analyses do not control for industry-fixed effects, they tend to find the relationship inconsistent across regions because of unobserved heterogeneity bias. Our current paper relaxes these assumptions and explores whether corruption control in a particular region affects the growth of small owner-managed farms differently than that of large organizational structures within the fixed livestock industry.

This study makes the most direct contribution to the literature on the effect of corruption control on livestock production. Using a unique dataset on registered corruption crime rates at the level of subnational units in Kazakhstan and Russia for 2010–2019, we analyze whether corruption control influences productive activities non-uniformly, depending on the producer's organizational structure. The rest of this paper is structured as follows: The next section provides background information about corruption and beef production in the two countries. Section 3 presents the conceptual framework. The data and the empirical approach will be introduced in Sections 4 and 5. Finally, we will provide the results, discuss their implications and limitations and end with a conclusion.

2 | BACKGROUND

2.1 | Corruption in Kazakhstan and Russia

Kazakhstan and Russia inherited corruption from the Soviet era. In the Soviet Union, corruption was rooted in the contradictions of an over-centralized economy. When the Soviet system of the government collapsed, but a replacement had not yet been established, corruption became an emerging mechanism of combining an emerging market system with bureaucratic barriers and personal enrichment (Rigi, 2017). Corruption acted as a compensation instrument, accelerating the decision-making process without clear legal standards and ineffective contract enforcement. Nowadays, in Kazakhstan and Russia, corruption is seen as a means of extracting rent and ensuring the loyalty of subordinates in the administrative hierarchy (Satpayev, 2014; Schulze et al., 2016). Corruption is entrenched in political and bureaucratic realms, yet higher-level corruption is not systematically sanctioned, as elites have practical immunity from prosecution (Janenova & Knox, 2020; Schulze & Zakharov, 2018). The legal framework governing corruption in independent Kazakhstan and Russia has been weakly consolidated. The anti-corruption legislation in Kazakhstan and Russia focuses on individualistic corruption cases with only a limited number of companies being held responsible for their involvement in corrupt activities. The prosecution processes rely on direct evidence but fail to fully utilize financial investigations to uncover and penalize instances of corruption (OECD, 2016).

Over the past decade, Kazakhstan and Russia attempted different measures to control corruption. Anti-corruption initiatives include prevention and control efforts such as stricter laws, digitalization, public accountability and individualistic prosecution of high-profile offenders (Janenova & Knox, 2020; Schulze & Zakharov, 2018). However, the development of both countries in the most popular composite corruption indices suggests a differing success of these attempts. While Kazakhstan improved its position during the period analyzed in this study, Russia's ranking does not show a substantial change. More specifically, Kazakhstan's percentile

rank in the World Bank's Worldwide Governance Indicator (WGI) increased from 17.54 in 2011 to 45.24 in 2019. Transparency International's Corruption Perception Index (CPI) indicates a similar development, the index increased from 27 to 34 out of 100.¹ Russia's percentile rank, according to the WGI, increased from 15.64 in 2011 to 23.81 in 2019. The CPI increased from 24 to 28. Observers claim that the mechanisms of the regulatory implementation of less bureaucratic procedures within public administration are still weak in both countries (OECD, 2020).

2.2 | Overview of cattle and beef production

As in many other post-Soviet economics, livestock underpins the livelihoods of large parts of rural households as a supplementary income or a means of self-sufficiency.² In Kazakhstan and Russia, the doctrine of economic diversification and the goal of ensuring national food security aimed at promoting concentration and professionalization of livestock production via subsidies and, at least in the case of Russia, selective import bans towards pork and dairy products (Kvartiuk & Herzfeld, 2021; Oshakbayev & Bozayeva, 2019; Pomfret, 2016). Cattle farming is progressively being transformed into higher standards and quality production, necessitating more sophisticated processing along export-oriented value chains (Petrick et al., 2018), yet most cattle are raised by family farms and small rural households with limited resources (Herzfeld, 2024; Robinson et al., 2021). Cattle farming also has garnered considerable attention in both countries due to its environmental impact, including the blue water footprint, desertification and greenhouse gas emissions (Alimaev et al., 2008; Bityukova & Borovikov, 2021). Thus, assessing the role of the institutional roots of good governance in determining cattle production growth is crucial, not just from an economic standpoint but also in terms of its environmental and social implications.

Looking closer at the evolution of cattle and beef production over the past 3 decades, it resembles a roller coaster. After the dissolution of the Soviet Union, large-scale dairy and beef production by formerly collective and state farms virtually ceased. Most state enterprises either sold and butchered their livestock or distributed it to independent farmers (Anderson & Swinnen, 2008; ur-Rahim et al., 2014). Rural households continued to raise the few cows and calves they used to have as a side activity besides wage work on the collective farms. As a result, Kazakhstan's beef and cattle production output dropped annually between 1990 and 1999. The sector started to recover in 2000 but did not achieve the Soviet-era level so far (Figure 1). Between 1990 and 1999, Russia saw a sharp decline in beef and cattle production (Figure 2). Since 1999, the declining trend has continued but at a slower pace.

Small livestock producers in both countries are among the most vulnerable agricultural producers. They have limited choices about where to keep their cows, but larger producers are more privileged in access to land and investment subsidies. Small households mainly stock around their settlements, resulting in environmental degradation and diminished livestock performance (Alimaev et al., 2008; Bityukova & Borovikov, 2021). The regulatory hurdles, such as minimum herd count and live weight thresholds, eliminate smaller producers who do not match the scale requirements of state support. Such discrimination creates an unprecedented competitive advantage for large enterprises over the rest of the agricultural sector and facilitates rent-seeking opportunities when obtaining permits and proof documentation required to receive subsidies (Robinson et al., 2021; Uzun et al., 2019).

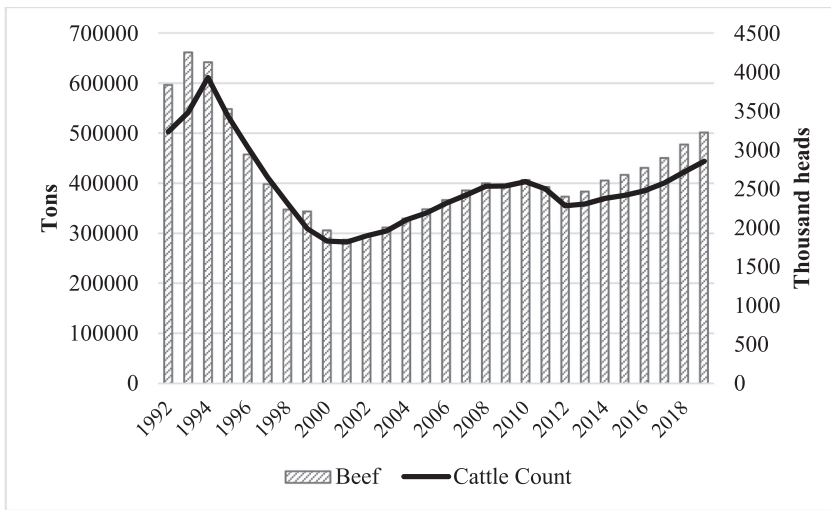


FIGURE 1 Beef production and cattle headcount, Kazakhstan 1992–2019. Source: FAOSTAT.

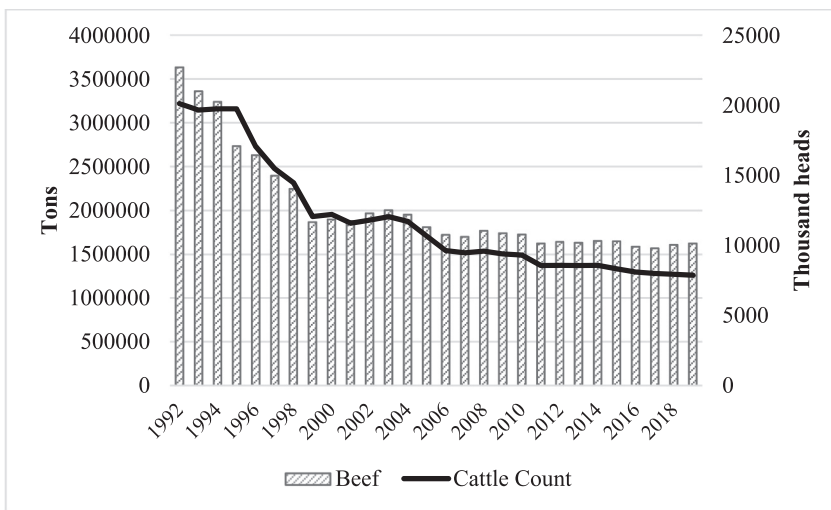


FIGURE 2 Beef production and cattle headcount, Russia 1992–2019. Source: FAOSTAT.

The agricultural support system is discriminatory not only between farm types but even between agricultural enterprises themselves. Only a few large farmers might receive large subsidies, while the majority receive just little stimuli (Petrick & Götz, 2019; Robinson, 2020). In 2016, for instance, Bryansk Meat Packers, a subsidiary of the Miaratorg agroholding, received 90.7% of all beef development subsidies, while other subsidized credits were allocated to only three firms (MinAgRus, 2017). In conjunction with anecdotal evidence, such sentiments support the hypothesis that corrupted activity benefits more adapted farm organizations while retarding development for vulnerable farmers. This discrepancy may call for specific institutional provisions and rules for policy coordination that may reduce the conflicts when redistributing public goods and services.

3 | CONCEPTUAL FRAMEWORK

The conceptual framework of this study explores the intricate relationship between anti-corruption efforts and productive activities, particularly within the context of the Kazakh and Russian cattle sectors. Existing literature presents mixed evidence regarding the impact of corruption on firm productivity (Gamberoni et al., 2016; Wu et al., 2017), making the effect of anti-corruption campaigns an empirical question subject to the dominance of either rent-seeking or lubricant theory.

According to the rent-seeking theory, corruption can significantly reduce firm productivity by redirecting revenues and production resources away from the companies (Kong et al., 2020). However, the impact of corruption goes beyond just decreased production activities. The unpredictable institutional environment, characterized by opportunistic actions and discretionary bureaucracy, leads to higher transaction costs for conducting business (Duvanova, 2014). Consequently, entrepreneurs experience greater uncertainty regarding future incomes, which distorts the incentives for innovative production and productive investments (Del Monte & Papagni, 2001).

The lubricant theory posits that corruption can enhance efficiency, address market failures and expedite decision-making in inadequately regulated environments (Guriev, 2004; Satpayev, 2014). This concept was especially relevant during the initial transition phase of post-Soviet countries, which lacked clearly defined legal standards (Schulze & Zakharov, 2018). Other examples show that bribery can enhance the effectiveness of long-term contracts between firms and the bureaucratic administration, positively impacting the firm's output and productivity growth (Vial & Hanoteau, 2010). Therefore, efforts to combat corruption in poorly regulated environments with ineffective institutions may have unintended consequences, potentially hindering production activities for agents that benefit from poorly regulated environments.

The costs of corruption, regulatory red tape and compliance impact small and large producers differently. Small producers often lack the financial resources and political connections to navigate corrupt systems effectively. They are more vulnerable to the extortion of informal payments for basic permits, approvals or market access (Robinson et al., 2021; Uzun et al., 2019). An uncertain business climate creates significant barriers to entry and growth for small producers, exacerbating inequality within the sector (Fisman & Svensson, 2007). On the other hand, large producers, with more substantial resources and better access to networks of influence, can often bypass costly regulatory processes through bribes or informal agreements. A corrupt environment allows large players to gain competitive advantages, such as quicker access to subsidies, favourable treatment in resource allocation and expedited administrative procedures (Kvartiuk & Herzfeld, 2023; Olken & Pande, 2012). Consequently, corruption can entrench the dominance of large producers, stifle competition and hinder overall sectoral growth by preventing more efficient small producers from expanding.

The legal criminalization of corruption in Kazakhstan and Russia in the 2000s brought substantial changes in the business landscape, leading to a reduction in opportunistic behaviour among officials and the restoration of the effectiveness of formal institutions across various regions (OECD, 2020). As a result, for the agricultural sector, it is reasonable to assume that farmers in areas with stringent anti-corruption measures may encounter fewer instances of

arbitrary bureaucracy. Specifically, the farmers are likely to benefit from a more transparent allocation of agricultural support, improved access to natural resources, simplified permission processes and, overall, achieve higher growth rates than farmers in regions with less stringent anti-corruption measures. Therefore, we hypothesize that anti-corruption efforts positively affect cattle production growth (Hypothesis 1).

Corruption can affect organizational structures in various ways, considering supply and demand (Goel & Nelson, 1998). On the supply side, farmers may offer inappropriate incentives for quick service or permit approval. On the demand side, authorities regulating livestock markets may abuse their power to extort informal benefits by delaying permission release. This interplay between supply and demand often results in a tendency to favour larger organizational structures and large projects, as these are perceived as lower-risk avenues for maximizing personal gain while minimizing detection (Del Monte & Papagni, 2001). Such favouritism allows larger agricultural producers to bypass or expedite costly regulatory compliance, giving them a competitive advantage over smaller producers who avoid corruption. In regions with stricter anti-corruption measures, administrations responsible for subsidy distribution, funding oversight and permission granting are expected to support all farmers equally, regardless of organizational size. Therefore, we hypothesize that in areas with more vigorous anti-corruption efforts, the growth among different organizational structures will be more balanced compared to regions with weak corruption control (Hypothesis 2).

The dynamic distribution of output between different organizational forms varies significantly due to how they allocate resources between productive activities, such as innovative production, and unproductive activities, like rent-seeking (Baumol, 1996). In an institutional environment that tolerates unproductive activities, firms with influential connections or informal incentives for bureaucrats may thrive, promoting corrupt practices (Lambsdorff, 2003). These practices, where personal connections are valued over qualifications, create uncertainty about future income and reduce motivation to invest in innovative production. The biased selection and the lack of incentives for productive innovations have long-term effects, resulting in a significant opportunity cost for future growth. We expect experienced and skilled farmers in regions with stronger anti-corruption measures to engage in organizational and technological innovations in cattle production. We hypothesize that anti-corruption measures are linked dynamically to the expansion of cattle production, yielding positive long-term impacts (Hypothesis 3).

Further, we link anti-corruption to the value-added production. In animal production, value-added production entails various regulatory stages, which can be managed within a farm or in specialized locations (Hobbs, 1996). Enforcing these regulations involves interactions between farms and public authorities, including activities like animal registration, inspection and monitoring to guarantee veterinary and food quality standards. However, farmers may circumvent quality requirements for value-added production, and regulatory bodies may not consistently and fairly enforce the rules (Petrick et al., 2014). Corrupt officials may expedite legitimate requests or disregard non-compliance in exchange for favours. Yet controlling corruption may reduce transaction costs in value-added production associated with opportunistic behaviour. We aim to assess whether increased anti-corruption measures have more impact on higher value-added products, like beef production growth, than the lower value-added products, such as cattle headcount growth (Hypothesis 4).

4 | DATA

Our dataset encompasses data from 73 regions in Russia and 13 regions in Kazakhstan, spanning from 2011 to 2019.³ We obtained cattle and beef production rates, land distribution, as well as labour data from the National Statistical and Ministry of Agricultural offices. Registered corruption cases are taken from the Ministries of Internal Affairs. The expenditures on agricultural services and support were obtained from Ministries of Finance, while loan rates for agriculture were gathered from the Central banks of the countries.

We distinguish three cattle and beef producer types: enterprises, peasant farmers and households. The enterprise farm type represents corporate farms of various legal forms, commonly associated with large farm organizations, though this is not always the case of scale. Among them are business partnerships, production cooperatives, unitary businesses and subsidiary plots. Peasant farms typically refer to middle-sized farms. These farms refer to legal entities created by an individual or a group of individuals who share ownership and carry out production jointly or are related by kinship. The enterprises and peasant farms primarily focus on commercial aims. Household farms are the personal subsidiaries and other personal farms of rural and urban settlements (Mussayeva, 2019; Rosstat, 2021). Unlike enterprises and peasant farms, household farms are exempt from mandatory legal registration and taxation (Hajdu et al., 2021).

The main cattle producers in Russia are enterprises and households, while in Kazakhstan, peasant farms and households hold the majority of the cattle herd size (Table 1). In Russia, enterprises produce more beef than peasant farmers, unlike in Kazakhstan. However, in both countries, households remain the main source of beef production. Upon closer examination of the dynamics among producers, it becomes evident that farmers of all types encounter regional fluctuations in herd size and beef production: All farm types are characterized by higher regional variation than variation over time in herd size and beef produced. The disparity at regional and country level dynamics suggests that the cattle and beef industry in these countries is structured differently with varying factors contributing to the distribution of cattle ownership and level of beef production.

Understanding the regional disparity also involves acknowledging the endemic challenge of corruption, a factor that can significantly impact agricultural business (Herzfeld et al., 2018). By definition, measuring the extent of corruption faces challenges. Various scholars propose different measures for operationalization. While subjective indicators rely on the assessment of respondents regarding their own experience of being asked for bribes, objective indicators rely on secondary statistics. Examples of objective indicators encompass the UN Survey of Crime Trends, the propensity of diplomats from different countries to break the traffic law, the number of convictions (Del Monte & Papagni, 2001; Hamilton & Hammer, 2018) or open cases of corruption crime rates. Cross-country index-based analysis often treats corruption uniformly within a country and is highly criticized (Schulze & Zakharov, 2018; Shirley, 2013).

To address these limitations, our analysis employs registered corruption cases as a proxy for anti-corruption efforts. The registered corruption cases include those charged with corruption, bribery, extortion and abuse of public power. Examples of corruption crimes include public officials accepting unofficial rewards to accelerate their duties, offering preferential treatment in granting government contracts, and citizens who pay bribes to get leniency in punishment, bypassing law enforcement or expediting bureaucratic operations. The registered corruption cases are more representative of the actual anti-corruption levels compared to the conviction-based data traditionally used in the empirical literature. While the number of prosecutions

TABLE 1 Descriptive statistics of cattle number and beef produced by countries from 2011 to 2019.

		St. dev.		
Variable	Mean	Overall	Between	Within
<i>Cattle number, thous. heads</i>				
Kazakhstan				
Enterprises	40,836.3	35,950.9	34,311.9	14,012.4
Peasant farms	136,851.0	115,098.2	106,701.2	51,455.2
Households	304,589.6	180,389.7	166,096.7	82,794.7
Russia				
Enterprises	118,868.1	119,233.4	118,038.5	21,179.4
Peasant farms	31,711.3	41,038.4	40,268.7	9074.8
Households	121,031.9	140,851.4	140,459.0	17,664.7
<i>Beef produced, thous. tons</i>				
Kazakhstan				
Enterprises	2490.1	2685.6	2111.3	1750.0
Peasant farms	6135.0	6489.4	6077.5	2779.2
Households	24,035.2	13,541.0	13,920.1	1718.1
Russia				
Enterprises	7405.9	8318.4	7940.8	2628.6
Peasant farms	1737.4	2185.5	2050.9	788.4
Households	12,729.8	14,554.6	14,498.3	2048.7

might be influenced by the quality of the courts, registered cases also capture unjustified accusations, making them a more comprehensive measure (e.g. Cole et al., 2009; Goel & Nelson, 1998).

Additionally, conviction rates may be more distorted than registered cases due to the ease of prolonging and sabotaging investigations. Failing to register a crime is easily proven, making it less likely to go unreported. Convictions refer to the number of individuals convicted, regardless of the number of crime cases. In contrast, registered corruption cases correspond to each detected criminal act, providing a better representation of the willingness to report corruption incidents and measuring the frequency of anti-corruption efforts. Additionally, registering a corruption incident case takes less time than criminal proceedings and convictions, which can span several years. Therefore, the relationship between registered cases and the timing of investment decisions at the farm level may be closer than that of conviction rates (Schulze et al., 2016; Zakharov, 2019).

In Kazakhstan and Russia, despite a prevailing centralized authority and legal systems, the incidence of corruption represented by registered cases varies from region to region. The higher within variation of registered cases per capita suggests a rather dynamic pattern of this variable over time. Based on official statistics, it can be seen that Kazakhstan recorded an average of 89.5 cases of registered corruption crimes over the 2011–2019 period, while Russia recorded 130.3

cases (Table 2). Based on the data, it appears that no region is free from reported incidents of corruption. To ensure that the overall number of registered corruption crimes is not affected by a region's size, we normalize the registered cases by population (Solon et al., 2015). By doing so, we find that both countries have a comparable level of registered corruption cases per capita. Our identification strategy rests on the assumption that the likelihood of registering a corrupt incident is constant across all regions in both countries. Thus, the variation in the measure of registered cases truly reflects higher efforts to combat corruption.

5 | EMPIRICAL FRAMEWORK

In the empirical framework, we aim to explore the relationship between anti-corruption efforts and the growth of cattle herd size and beef production across various organizational structures of farms at the regional level in Kazakhstan and Russia. We analyze how factors such as input and output prices, agricultural support, land distribution and registered corruption incidents influence production growth over time. We assume that regions with stronger anti-corruption efforts foster better performance in animal raising and meat production by encouraging investments and ensuring inclusive access to resources and services. In the long-term nature of investments for cattle production—where cattle fattening extends beyond a year, and cows are milked over several years—a dynamic model is essential. The dynamic model approach allows us to estimate both the short- and long-term effects of anti-corruption efforts on farm performance thereby assisting in testing the research hypotheses. Given the importance of anti-corruption efforts, we anticipate that the estimated coefficient of anti-corruption efforts on output growth will be positive, reflecting the inertia of corruption control. The primary justification for including the anti-corruption efforts variable is that regions with historically high rates of reporting corruption crime incidences are less likely to misallocate or misappropriate state-funded transfers because of a higher probability of punishment.⁴ When integrating the corruption control parameter in the panel model specification, we can at least make the following observation: If two regions have identical registered corruption crime rates in the past, and all other factors are held constant, then the estimated coefficient captures the relationship between anti-corruption effort and production growth (Wooldridge, 2012).

The data description reveals the dynamic development of cattle and beef production growth, which varies not only across regions but also among different organizational structures of farms. To further elucidate the impact of anti-corruption efforts on production growth, the following equation has been developed to model the dynamic relationship between anti-corruption measures and the growth in cattle herd size (Windmeijer, 2005):

$$\ln\left(\frac{Y_{i,t}^f}{Y_{i,t-1}^f}\right) = \alpha_0^f + \alpha_1^f \ln\left(Y_{i,t-1}^f\right) + \beta_0^f AC_{i,t} + \beta_1^f AC_{i,t} \ln\left(Y_{i,t-1}^f\right) + \beta_2^f AC_{i,t-1} + \beta_3^f AC_{i,t-1} \ln\left(Y_{i,t-1}^f\right) + \sum_{j=0,\dots,m} \gamma_j^f Z_{i,t} + u_i^f + \mu_t^f + \varepsilon_{i,t}^f, \quad (1)$$

In this equation, $Y_{i,t}^f$ and $Y_{i,t-1}^f$ represent cattle herd size for farm type f , in region i and at a period t ; $AC_{i,t}$ denotes anti-corruption efforts represented by registered cases of corruption; $Z_{i,t}$

TABLE 2 Descriptive statistics of registered corruption cases by countries from 2011 to 2019.

Variable	Mean	St. dev.		
		Overall	Between	Within
Kazakhstan				
Registered corruption cases (per 100,000 population)	8.76	5.43	3.43	4.30
Registered corruption cases (total)	89.53	53.83	31.66	44.33
Russia				
Registered corruption cases (per 100,000 population)	8.30	5.20	3.18	4.12
Registered corruption cases (total)	130.25	116.15	93.33	69.90

is a vector of control variables; u_i captures region-specific fixed effects; μ_t accounts for time-specific fixed effects; and $\varepsilon_{i,t}$ represents an error term.

Similarly, the relationship between anti-corruption efforts and beef production growth is expressed by the following equation:

$$\ln\left(\frac{V_{i,t}^f}{V_{i,t-1}^f}\right) = \delta_0^f + \delta_1^f \ln\left(V_{i,t-1}^f\right) + \varphi_0^f AC_{i,t} + \varphi_1^f AC_{i,t} \ln\left(V_{i,t-1}^f\right) + \varphi_2^f AC_{i,t-1} + \varphi_3^f AC_{i,t-1} \ln\left(V_{i,t-1}^f\right) + \sum_{j=0,\dots,m} \lambda_j^f Z_{i,t} + \omega_i^f + \tau_t^f + \epsilon_{i,t}^f, \tag{2}$$

Here, $V_{i,t}^f$ and $V_{i,t-1}^f$ represent beef production rates for farm type f , in region i and at a period t ; $AC_{i,t}$ denotes registered corruption cases; $Z_{i,t}$ is a vector of control variables; ω_i captures region-specific fixed effects; τ_t accounts for time-specific fixed effects; and $\epsilon_{i,t}$ represents an error term.

Taking the derivative from Equations (1) and (2) with respect to dynamic factors enables quantification of the growth rates for cattle herd size and beef production respectfully. The conditions $\partial \ln\left(Y_{i,t}^f/Y_{i,t-1}^f\right)/\partial_1 \ln\left(Y_{i,t-1}^f\right) < 0$ and $\partial \ln\left(V_{i,t}^f/V_{i,t-1}^f\right)/\partial_1 \ln\left(V_{i,t-1}^f\right) < 0$ denote a quicker growth or convergence within regions that produce the least.

Building on the specifications of Equations (1) and (2), the core relationships can be further explored through interaction terms between anti-corruption efforts and production levels for each farm type. These interaction terms reveal the dynamics of direction and magnitude of the anti-corruption effects on production growth across different farm types. Specifically, the short-run effects of anti-corruption efforts are captured by the estimated marginal effects of the anti-corruption variable from Equation (1) for cattle production and from Equation (2) for beef production. The derivation demonstrates that regions experiencing more anti-corruption efforts expressed by more frequent registered corruption cases tend to exhibit faster growth in both cattle herd size and beef production, as indicated by the conditions $\partial \ln\left(Y_{i,t}^f/Y_{i,t-1}^f\right)/\partial \ln(AC_{i,t}) > 0$ and $\partial \ln\left(V_{i,t}^f/V_{i,t-1}^f\right)/\partial \ln(AC_{i,t}) > 0$, respectively. The long-run effects (LRE) for cattle herd size growth can be derived by the following expression (Chudik et al., 2016):

$$LRE^f = \frac{\partial \ln(Y_{i,t}^f/Y_{i,t-1}^f) / \partial \ln(AC_{i,t}) + \partial \ln(Y_{i,t}^f/Y_{i,t-1}^f) / \partial \ln(AC_{i,t-1})}{1 - \partial \ln(Y_{i,t}^f/Y_{i,t-1}^f) / \partial \ln(Y_{i,t-1}^f)} \quad (3)$$

However, given the dynamic nature of the model, it may suffer from serial correlation, which raises the issue of endogeneity. Thus, even in fixed- or random-effects models, ordinary least squares estimates will be inconsistent, as the dynamic term (i.e. the lagged dependent variable) will be correlated with the unobserved panel-level effects (Arellano & Bond, 1991). The lagged variables in Equations (1) and (2) will be endogenous to the fixed effects in the error term.

As a strategy for the identification, we must equip the dynamic term with instruments that are exogenous to the region-level fixed effects to ease the problem (Blundell & Bond, 1998). Such issues can be handled by specifying a Generalized Method of Moments (GMM) estimator for the linear dynamic panel model (Arellano & Bond, 1991; Holtz-Eakin et al., 1988). The assumptions for system GMM estimation imply that instruments utilized as the lagged dependent variables among the regressors are uncorrelated with the fixed effects. As long as this assumption holds, the econometric model controls time-invariant unobserved effects, which allows for estimating the long-run effect of anti-corruption efforts in livestock production (Roodman, 2009). Lagged variables demonstrate weak properties for first differences in GMM models (Dithmer & Abdulai, 2017). Utilizing instruments that account for a small proportion of the variance in possibly endogenous explanatory variables might result in inefficient and biased coefficient estimations (Bound et al., 1995). To address this problem, we use a two-step System-GMM estimator that incorporates finite-sample standard error correction (Windmeijer, 2005).⁵ The production effects of varying degrees on cattle raising and beef production are embedded into the theoretical assumptions of the two-step GMM with finite-sample corrected standard errors, selected as the empirical estimation strategy for analytical purposes. To explore the validity of the identifying assumptions, we specify Arellano and Bond (1991) tests of autocorrelation and Hansen (1982) test of over-identifying restrictions.

Following the theoretical review, we postulate that regional cattle and beef output is determined by a set of characteristics presented in Table 3. The specification of right-hand variables resembles methods by Deller et al. (2003) and Petrick and Götz (2019). The measures are centred on capturing proxies for demand, such as output prices, and the capacity of the regional market to supply, such as input prices, labour machinery and area of agricultural land, as well as proxies for institutional support, such as credits, expenditure on agriculture, cooperation and privately owned land.

6 | RESULTS

Before moving to the estimation results, we look at how anti-corruption initiatives have evolved in relation to the cattle and beef production levels of different types of farms. Figure 3 illustrates the dynamics.⁶ The first observation depicts that total cattle and beef production change over time with anti-corruption efforts. For businesses run by enterprise farmers, there is a negative association between anti-corruption efforts and both cattle and beef production. Production levels for both commodities decline when corruption control measures are tightened. The relationship between anti-corruption efforts and cattle, as well as beef production, is less evident

TABLE 3 Descriptive statistics of independent variables over the 2011–2019 period.

Variable	Mean	Standard deviations		
		Overall	Between	Within
Registered corruption cases per capita	0.84	0.52	0.32	0.41
Expenditure on agriculture, mln. USD	56.28	55.11	48.03	27.46
Credits in agriculture, mln. USD	175.34	275.10	232.47	148.98
Price of inputs (cattle feed) per kg, USD	0.28	0.16	0.12	0.10
Price of cattle per head, USD	1551.22	668.27	524.52	426.26
Price of beef per kg, USD	5.49	1.47	0.64	1.33
Labour employed in agriculture, thous. people	19.70	18.44	18.22	3.42
Number of tractors, thous. units	7058.76	6924.00	6913.16	598.52
Number of cooperatives, thous. units	0.32	1.24	0.40	1.18
Area of agricultural land, ha	5076.38	6421.78	6173.35	1877.09
Area of privately owned land, ha	1.49	1.76	1.77	0.04

for peasant farmers. The cattle production run by households shows a closer connection between output and anti-corruption efforts. The basic correlation analysis depicts that anti-corruption efforts may closely relate to the production output of agricultural commodities, especially for enterprise producers and households. However, it is insufficient to base conclusions on comparing variables' means alone. Therefore, we employ dynamic panel data econometric specifications that consider the biases brought on by both observed and unobserved confounders.

The theory only gives a general knowledge of how corruption control connects to the dynamics of production progress; therefore, the lag length must typically be discovered empirically. Prior to the selection of the empirical model, we explore methodological issues such as the model and dynamic specifications to answer the research questions appropriately. We specify different econometric models with the restriction that each places on the dynamic model's parameters. We perform *t*-tests or *F*-tests to verify whether the partial adjustment or static specifications are consistent with the dynamic model, Equations (1) and (2).⁷ By doing this, we aim to investigate the possibility that coefficients on weakly exogenous variables may exhibit large directional variations. In addition, this process helps prevent biased estimations of lag lengths and explores short- and LRE.

The estimation results of Equations (1) and (2) applications to the regional panel dataset based on the dynamic model for the growth level of cattle headcount are shown in Figure 4, and the beef production growth rates are shown in Figure 5, respectively.⁸ Columns (2)–(4) detail the output for each producer type. These are then compared to the aggregate production growth across all producers in Column (1). The statistical calculations suggest that the variables jointly explain the cattle and beef production growth, which is revealed by *F*-statistics equalling between 4.29 ($p = 0.000$) and 10.36 ($p = 0.000$) for all organizational structures of farms.

In general, regions with greater financial inputs, as defined by agricultural expenditures and credit amounts, appear to demonstrate slightly better growth rates in cattle and beef production, as shown by the positive parameter discovered on the governance parameters. Particularly,

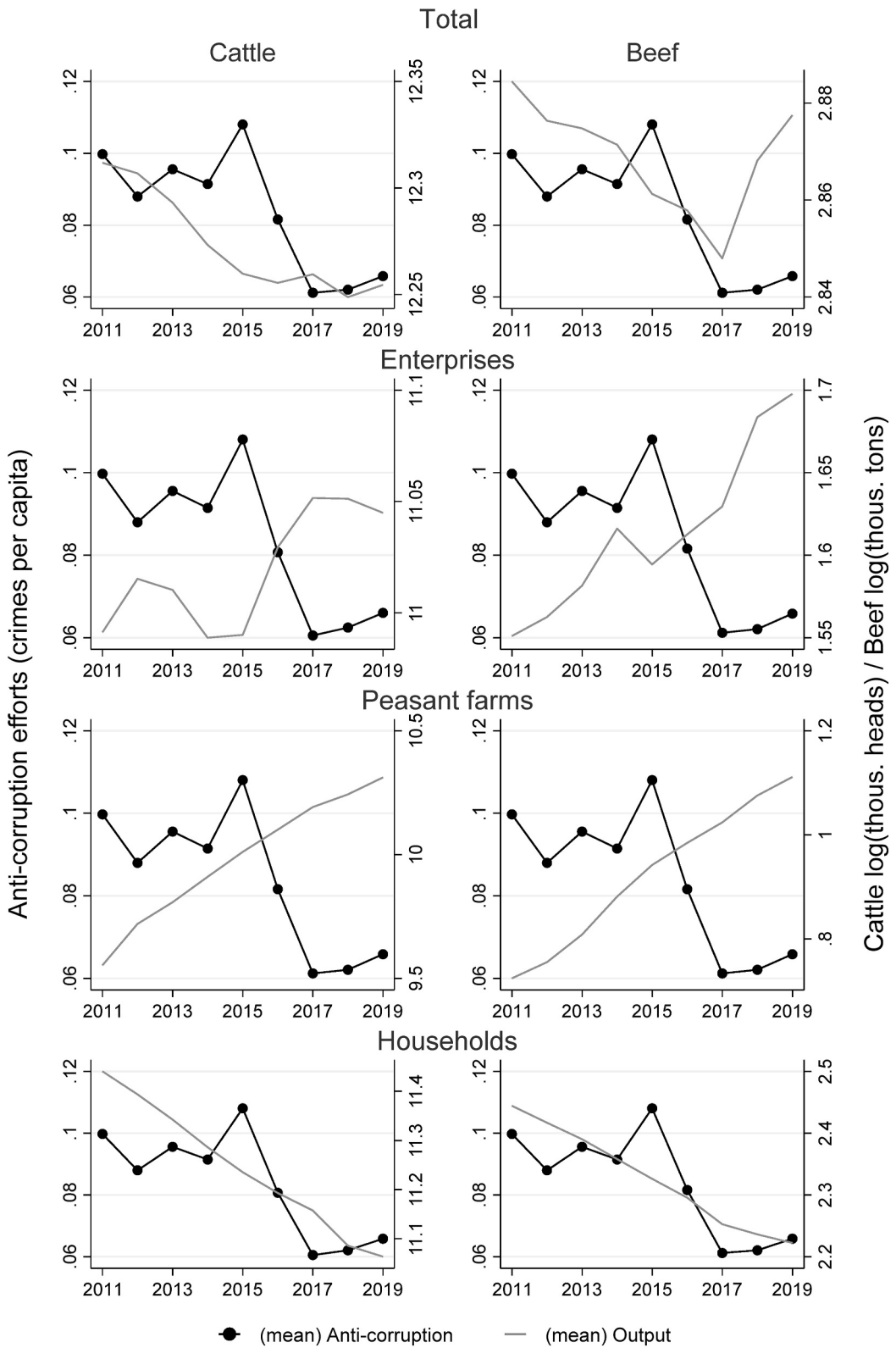


FIGURE 3 The relationship between the registered corruption cases and cattle headcount as well as the quantity of beef produced.

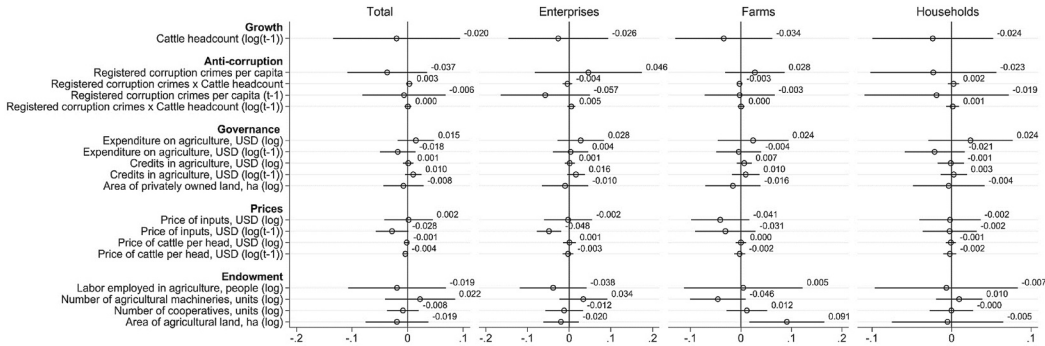


FIGURE 4 Estimates of cattle headcount growth across farm types.

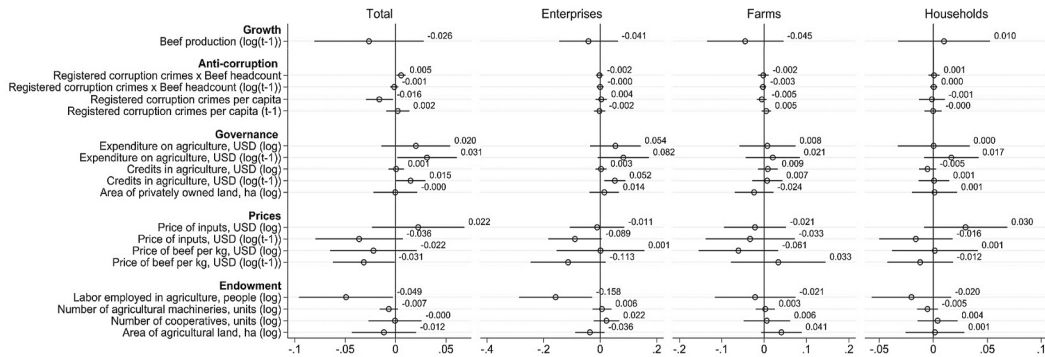


FIGURE 5 Estimates of beef production growth across farm type.

enterprises experience higher growth in cattle and beef production in regions where the state provides more support for agriculture. Peasant farmers witness better cattle and beef production growth in regions with more resources available for credits.

In regions with relatively higher input prices, cattle and beef production expand more slowly than in regions with lower input prices. Peasant farmers are the most sensitive to an increase in input prices compared to other organizational forms of farms. Negative parameters in the group of endowment factors suggest that in regions where more workers are employed in agriculture and have larger agricultural land areas, enterprises and households experience a slower growth of cattle and beef production. In contrast, peasant farmers tend to thrive in regions with abundant agricultural land, a high number of cooperatives and a strong agricultural labour force. Yet the enterprises increase production growth in cattle and beef in regions with higher levels of mechanization or a greater number of agricultural machinery.

One of the key findings directly relates to the production growth as quantified by marginal effects of the dynamic factor. The outcomes of these assessments are shown in the first rows of Table 4. We observe that the estimated marginal effects for lagged log cattle headcount and lagged log beef volumes are negative and less than one for enterprises and peasant farms. These estimates indicate that beef production and cattle herd size among enterprises and peasant farmers grow more rapidly in regions with lower production levels than those with larger beef volumes and cattle numbers. The larger coefficients for peasant farms indicate more rapid convergence across regions.

TABLE 4 Estimated marginal effects for growth rates and anti-corruption efforts.

Variables	(1) Total	(2) Enterprises	(3) Peasant farms	(4) Households
Cattle production				
$\partial \ln(Y_{i,t}^f/Y_{i,t-1}^f) / \partial \ln(Y_{i,t-1}^f)$				
Cattle number (log(t-1))	0.010 (0.051)	-0.016 (0.044)	-0.054 (0.034)	0.008 (0.031)
$\partial \ln(Y_{i,t}^f/Y_{i,t-1}^f) / \partial \ln(AC_{i,t})$				
Registered corruption crimes per capita	0.122 (0.205)	-0.036 (0.248)	0.111 (0.239)	0.469 (0.192)
Registered corruption crimes per capita (t-1)	-0.068 (0.183)	0.071 (0.228)	0.042 (0.169)	-0.350 (0.280)
Beef production				
$\partial \ln(V_{i,t}^f/V_{i,t-1}^f) / \partial \ln(V_{i,t-1}^f)$				
Beef production, tons (log(t-1))	0.010 (0.027)	-0.043 (0.041)	-0.084 (0.039)	0.026 (0.022)
$\partial \ln(V_{i,t}^f/V_{i,t-1}^f) / \partial \ln(AC_{i,t})$				
Registered corruption crimes per capita	-0.012 (0.108)	-0.213 (0.326)	-0.733 (0.455)	0.016 (0.098)
Registered corruption crimes per capita (t-1)	-0.198 (0.118)	-0.173 (0.302)	0.279 (0.387)	0.078 (0.083)

Note: Standard errors in parentheses.

The central findings of interest to the research hinge on the influence of anti-corruption efforts on rates of growth in cattle and beef production. As can be seen in Table 4, the link between the anti-corruption level and the cattle herd size is inconsistent among farm types. While the number of registered corruption cases appears to relate to an increase in cattle headcount growth, particularly for households and less so for peasant farmers, this relationship is inverse for enterprises. Although the increase in anti-corruption efforts has a negative temporary association with enterprises' cattle production growth, the cumulative impact of corruption turns out to be positive when we account for the estimated marginal effect of anti-corruption efforts in the preceding year. The positive sign of the lagged marginal effect of anti-corruption efforts on cattle herd size growth for peasant farmers demonstrates the increasing long-run effect of corruption control on cattle production growth. The positive sign of the contemporaneous effect and negative sign of the delayed effect, as measured by marginal effects of anti-corruption efforts on households' cattle production growth, indicate a positive relation between stronger corruption control and cattle herd size growth, albeit at diminishing rates in the long run. These results show that stronger corruption control contributes to an acceleration in the growth rate of cattle production, particularly, in regions where a greater proportion of cattle production is carried out by peasant farmers, as opposed to enterprises or household farms.

Notably, the connection between the anti-corruption level and upper-stream beef production is weaker than in lower-stream cattle production chains. This effect is observed by the magnitude of estimated marginal effects in Table 4. We observe that when the anti-corruption effort expands, downward pressure is placed on beef production growth for all types of producers except households. The reason for the diminished impact of anti-corruption measures on upper-stream beef output may be attributed to the majority of the supporting measures being directed towards upper-stream cattle production owing to pedigree herd revitalization strategies implemented in Kazakhstan and Russia (Kobayashi et al., 2007). Consequently, cattle production in the upper stream becomes more susceptible to resource misallocation and misappropriation as compared to upper stream beef production.

The effect of strengthening corruption control on production growth varies dynamically across different farm types. This discrepancy posits that in regions with stronger corruption control environments, peasant farmers grow faster than similar farmers in regions with weaker corruption control. For enterprises, anti-corruption efforts are associated with a negative immediate effect on cattle production growth, yet this relation is inverse in the long run, where corruption control is associated with increasing growth. For households, the impact of the initial efforts to combat corruption relates to faster growth, albeit at a diminishing rate in the long run. It is noteworthy that for commercially oriented farm structures such as enterprises and peasant farms, anti-corruption efforts have a positive long-term impact on cattle production growth. Whereas, for households engaged in subsistence farming and external income generation, anti-corruption efforts play a less significant role in herd size growth in the long run. In beef production, the anti-corruption efforts are associated with faster growth only for the household type of producers.

7 | ROBUSTNESS AND LIMITATIONS

7.1 | Robustness check

Finally, we examine the validity of our results, focusing on additional methodological concerns. First, we verify the specification of the econometric and dynamic specifications. This process helps prevent uncertainty over which lags are important and enables us to verify the lag lengths that are most likely to have a long-term effect. To address this, we employ the machine learning technique, least absolute shrinkage and selection operator (Lasso) to test different dynamic restrictions of anti-corruption impact on growth rates.⁹ The Lasso technique tends to shrink the coefficients of less important predictors exactly to zero, effectively performing variable selection. In the context of dynamic models, it can automatically select the most relevant lagged variables by setting the coefficients of less significant lags to zero (Hsu et al., 2008). The results are robust to partial adjustments specified in Table A1.

Next, we address the possibility of magnitude-direction variation between Kazakhstan and Russia by calculating separate effects for the anti-corruption effort within each country. To assess robustness, we utilize the same two-step System-GMM estimator that corrects for finite-sample standard errors as in the main analysis part. Figure A2 illustrates (APPENDIX A) the consistency between estimated coefficients. We see that the direction of the effect of anti-corruption efforts on cattle production growth is preserved across the countries, while the magnitude of the effect is larger in Russia. Our findings indicate that each key finding shown in Table 4 holds.

The second crucial robustness test is whether our estimates hold with a more parsimonious model, the fixed effects model. Specifically, we re-estimate the model using a different specification that includes country and year dummy variables. Adding time and country-fixed effects to a panel model with a lagged endogenous variable may result in inconsistent estimates and a considerable rise in multi-collinearity. However, excluding time and unit effects may cause bias when time series are short and the number of cross-sections approaches infinity (Kiviet, 1995). Results from these fixed effects models reported in Figure A3 (APPENDIX A) largely mirror those reported in Table 4.

7.2 | Limitations

One of the challenges associated with using secondary data sources is the accuracy of agricultural data, which has been criticized for its measurement accuracy. Nevertheless, Medvedev and Nefedova (2021) confidently affirm that the data on the livestock population is reliable and reflects the data reported by official statistical offices. Their research revealed a strong correlation between the number of livestock farm constructions displayed in remote sensing images and the number of livestock animals reported by state agencies.

Another concern stemming from the lack of available data is the use of cattle headcount growth as the dependent variable. A more unbiased estimator of livestock development would be livestock productivity instead of merely tracking herd size increase. This notion is supported by literature, which suggests that one of the primary objectives of farm restructuring and transitioning to global markets is to enhance productivity, such as increasing the weight of carcasses per cow, rather than just scaling up herd size (Liefert & Swinnen, 2002). However, herd size is the only traceable and reliable variable in secondary sources for measuring livestock progress. Other studies on former Soviet countries have relied on livestock herd size growth as a dependent variable and proxy for gauging livestock progress (Petrick & Götz, 2019). Nonetheless, having data that offers a more accurate measurement of livestock productivity would undoubtedly contribute to a more informed discussion on livestock development progress.

Another potential avenue for further exploration concerning anti-corruption efforts could be proposed in future research. The current data in the paper only covers low-level cases of corruption crimes, as the legislation in Russia and Kazakhstan focuses on individualistic corruption cases. Higher-level corruption, such as by elites or firms, is not systematically penalized (Janenova & Knox, 2020; Schulze & Zakharov, 2018). While this allows for identifying the effectiveness of a bottom-up approach against corruption in enabling livestock growth expansion, it also prevents defining the top-down effect of corruption control. Further research on corruption control, with top-level corruption data, could enrich ongoing debates on which approach is more effective and explore how combining bottom-up and top-down approaches can foster sustainable agricultural development.

Given the complexities and limitations of using secondary data, it is crucial to consider the broader methodological implications for research in this area. While accurate data is fundamental, the choice of analytical models also plays a significant role in the reliability of the findings. For instance, while lagged variables can be useful instruments, their applications in GMM models can introduce issues that are less pronounced in Instrumental Variables models. Specifically, problems with instrument validity, autocorrelation and finite sample biases can make GMM less reliable when using self-lagged instruments. Instrumental Variables models, due to their simpler structure and fewer instruments, might handle these issues more robustly,

making them preferable in situations where the weaknesses of self-lagged instruments are likely to manifest (Hayakawa, 2016; Kiviet, 2009).

8 | CONCLUSION

This study investigates how differences in anti-corruption efforts influence progress in cattle and beef production growth in Kazakhstan and Russia. We found that control of corruption has a mixed influence on production rates across different farm types, showing a more positive influence on commercially oriented organizational structures. The data analysis demonstrates that in regions with stronger corruption control, peasant farmers, referred to as medium-sized producers, grow on average faster than other agribusiness structures in regions with weaker corruption control. Enterprises, the larger producers, produce less in regions with higher anti-corruption levels, but the cumulative dynamic effect of corruption control is associated with increased production in the long run. The smallest household producers, primarily focused on self-sufficiency and external income generation, have a favourable relationship with anti-corruption initiatives, but this relationship diminishes over time. Unlike our expectations, the connection between the anti-corruption level and upper-stream beef production is weaker than in lower-stream cattle production chains.

This study makes the most direct contribution to the existing literature on the institutional effects on agricultural activity. It underscores the crucial role of the institutional environment in the development of the agricultural sector. Our research findings reveal a novel perspective: In countries with highly centralized authorities where corruption is systematic, larger organizational structures of agribusiness may benefit from a less controlled and transparent environment in the short run. Conversely, in an institutional environment with more anti-corruption efforts, not only large but also small and medium-sized organizational structures are likely to experience business progress in the long run.

Second, we make a more general contribution to the literature examining the effects of corruption control on production growth. We believe that an important contribution of this paper is highlighting the divisibility of the effect of corruption control on development progress. Inconsistencies among development economists are that corruption either may grease or sand the wheel of development progress. Due to such inconsistency, researchers have often assumed the direction of the influence of corruption to be homogeneous when studying the effect of the changes in the economic environment on sustainable development, trade, production or investment constrained by corruption control arrangements. We infer that the relationship between productivity and corruption is non-linear and differs across business organizational structures.

Thirdly, we underscore the practical implications of our findings. The dynamics of the effect is crucial, as the contemporaneous effect of efforts to combat corruption may differ significantly from the cumulative delayed effect. Our findings propose important suggestions that regions that make more effort to control corruption create a more favourable environment for agribusiness growth. In countries with transition economies, such as Kazakhstan and Russia, adopting effective methods to control corruption has a positive long-run impact despite potential short-run declines in production within certain organizational structures. Therefore, implementing policies to combat corruption and enhance the effectiveness of local public institutions could significantly boost production growth in the long run.

ACKNOWLEDGEMENTS

We express our sincere gratitude to Professor Michael Koetter and the two anonymous reviewers for their invaluable feedback and constructive suggestions. We appreciate their dedicated time and effort.

Open Access funding enabled and organized by Projekt DEAL.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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ENDNOTES

- ¹ The CPI in 2011 has been scaled between 0 and 10. Index values were 2.7 for Kazakhstan and 2.4 for Russia.
- ² According to National Statistic Offices (2019), 42% of the population in Kazakhstan and 25% of the population in Russia reside in rural areas.
- ³ We exclude certain regions for geographical reasons: for example, the Mangistau region in Kazakhstan, which is located in a desert zone; and the Kamchatka Krai in Russia's East-North, which is located in a freezing area. Additionally, we omit cities of administrative significance, which have the same status as regions but play only a minor role in livestock production. Finally, the occupied territories of Ukraine have been excluded.
- ⁴ The probability of punishment is the most powerful deterrent against corruption. The likelihood that a corrupt public official would be detected, investigated, detained, convicted and eventually sentenced is extremely important. If the likelihood of corruption being punished is low, even the harshest punishment cannot effectively deter it (Wang, 2018).
- ⁵ The finite-sample correction to the two-step covariance matrix derived by Windmeijer (2005) is available in the STATA package xtbon2 (Roodman, 2009).
- ⁶ We acknowledge that cases of registered corruption dropped between 2015 and 2017. However, addressing the reasons behind this decline is beyond the scope of this paper. Further research might be required to develop an appropriate explanation.
- ⁷ The table with specification analysis is presented in Table A1 of APPENDIX A.
- ⁸ The estimated coefficients with standard errors, Arellano–Bond test for autocorrelation and Hansen test of overidentification restrictions are also presented in Table A2 and Table A3 of APPENDIX A.
- ⁹ Figure A1, APPENDIX A, provides the visual representation of the path of each coefficient on the lagged value over the search grid for the lasso penalty parameters as well as Table A4, APPENDIX A, displays a table showing the selected lags after lasso estimation results. All estimations are made in STATA using the “lasso” command.

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How to cite this article: Rakhmetov, R., & Herzfeld, T. (2025). Do bigger farms suffer less from corruption? Anti-corruption efforts and the recovery of livestock production. *Economics of Transition and Institutional Change*, 33(3), 553–581. <https://doi.org/10.1111/ecot.12439>

APPENDIX A

TABLE A1 Restrictions of the dynamic model.

Model	Restriction
$Y_{i,t} = \alpha_0 + \alpha_1 Y_{i,t-1} + \beta_0 AC_{i,t} + \beta_1 AC_{i,t-1} + \sum \beta_j Z_{i,t} + \theta_{i,t}$	None
$Y_{i,t} = \alpha_0 + \alpha_1 Y_{i,t-1} + \beta_0 AC_{i,t} + \sum \beta_j Z_{i,t} + \theta_{i,t}$	$\beta_1 = 0$
$Y_{i,t} = \alpha_0 + \beta_0 AC_{i,t} + \sum \beta_j Z_{i,t} + \theta_{i,t}$	$\alpha_1 = \beta_1 = 0$
$Y_{i,t} = \alpha_0 + \beta_0 AC_{i,t} + \beta_1 AC_{i,t-1} + \sum \beta_j Z_{i,t} + \theta_{i,t}$	$\alpha_1 = 0$
$Y_{i,t} = \alpha_0 + \alpha_1 Y_{i,t-1} + \beta_1 AC_{i,t-1} + \sum \beta_j Z_{i,t} + \theta_{i,t}$	$\beta_0 = 0$

* $\theta_{i,t} = u_i + \mu_t + \varepsilon_{i,t}$.

TABLE A2 Estimates of the cattle headcount across farm types.

Variables	(1) Total	(2) Enterprises	(3) Peasant farms	(4) Households
Cattle headcount (log(t-1))	-0.020 (0.057)	-0.026 (0.060)	-0.034 (0.048)	-0.024 (0.038)
Registered corruption crimes per capita	-0.037 (0.036)	0.046 (0.065)	0.028 (0.030)	-0.023 (0.040)
Registered corruption crimes x Cattle headcount (log(t-1))	0.003 (0.003)	-0.004 (0.006)	-0.003 (0.003)	0.002 (0.004)
Registered corruption crimes per capita (t-1)	-0.006 (0.038)	-0.057 (0.054)	-0.003 (0.035)	-0.019 (0.046)
Registered corruption crimes (t-1) x Cattle headcount (log(t-1))	0.000 (0.003)	0.005 (0.005)	0.000 (0.004)	0.001 (0.004)
Expenditure on agriculture, USD (log)	0.015 (0.017)	0.028 (0.028)	0.024 (0.035)	0.024 (0.027)
Expenditure on agriculture, USD (log(t-1))	-0.018 (0.016)	0.004 (0.021)	-0.004 (0.022)	-0.021 (0.019)
Credits in agriculture, USD (log)	0.001 (0.005)	0.001 (0.006)	0.007 (0.007)	-0.001 (0.008)
Credits in agriculture, USD (log(t-1))	0.010 (0.007)	0.016 (0.011)	0.010 (0.014)	0.003 (0.008)
Area of privately owned land, ha (log)	-0.008 (0.018)	-0.010 (0.028)	-0.016 (0.027)	-0.004 (0.023)
Price of inputs, USD (log)	0.002 (0.022)	-0.002 (0.029)	-0.041 (0.029)	-0.002 (0.019)
Price of inputs, USD (log(t-1))	-0.028 (0.015)	-0.048 (0.015)	-0.031 (0.030)	-0.002 (0.017)
Price of cattle per head, USD (log)	-0.001 (0.002)	0.001 (0.008)	0.000 (0.005)	-0.001 (0.003)
Price of cattle per head, USD (log(t-1))	-0.004 (0.002)	-0.003 (0.007)	-0.002 (0.005)	-0.002 (0.004)
Labour employed in agriculture, people (log)	-0.019 (0.044)	-0.038 (0.040)	0.005 (0.059)	-0.007 (0.045)
Number of agricultural machineries, units (log)	0.022 (0.032)	0.034 (0.029)	-0.046 (0.028)	0.010 (0.015)

(Continues)

TABLE A2 (Continued)

Variables	(1) Total	(2) Enterprises	(3) Peasant farms	(4) Households
Number of cooperatives, units (log)	−0.008 (0.014)	−0.012 (0.023)	0.012 (0.020)	−0.000 (0.014)
Area of agricultural land, ha (log)	−0.019 (0.028)	−0.020 (0.021)	0.091 (0.037)	−0.005 (0.035)
Constant	0.342 (0.450)	−0.305 (0.598)	−0.389 (0.678)	0.175 (0.382)
Observations	603	601	603	601
Number of id	84	83	84	83
First-order autocorrelation	0.0906	0.0123	0.00404	0.0729
Second-order autocorrelation	0.659	0.585	0.112	0.271
Hansen over ID test	0.401	0.580	0.205	0.120

Note: Standard errors in parentheses.

TABLE A3 Estimates of beef production across farm types.

Variables	(1) Total	(2) Enterprises	(3) Peasant farms	(4) Households
Beef production (log(t-1))	−0.026 (0.027)	−0.041 (0.052)	−0.045 (0.045)	0.010 (0.021)
Registered corruption crimes per capita	0.005 (0.002)	−0.002 (0.006)	−0.002 (0.006)	0.001 (0.003)
Registered corruption crimes x Cattle headcount (log(t-1))	−0.001 (0.002)	−0.000 (0.006)	−0.003 (0.004)	0.000 (0.002)
Registered corruption crimes per capita (t-1)	−0.016 (0.007)	0.004 (0.010)	−0.005 (0.006)	−0.001 (0.006)
Registered corruption crimes (t-1) x Cattle headcount (log(t-1))	0.002 (0.006)	−0.002 (0.010)	0.005 (0.006)	−0.000 (0.004)
Expenditure on agriculture, USD (log)	0.020 (0.017)	0.054 (0.045)	0.008 (0.033)	0.000 (0.017)
Expenditure on agriculture, USD (log(t-1))	0.031 (0.015)	0.082 (0.046)	0.021 (0.032)	0.017 (0.013)
Credits in agriculture, USD (log)	0.001	0.003	0.009	−0.005

TABLE A3 (Continued)

Variables	(1) Total	(2) Enterprises	(3) Peasant farms	(4) Households
	(0.004)	(0.010)	(0.012)	(0.004)
Credits in agriculture, USD (log(t-1))	0.015 (0.007)	0.052 (0.019)	0.007 (0.018)	0.001 (0.007)
Area of privately owned land, ha (log)	-0.000 (0.011)	0.014 (0.026)	-0.024 (0.023)	0.001 (0.011)
Price of inputs, USD (log)	0.022 (0.023)	-0.011 (0.048)	-0.021 (0.037)	0.030 (0.019)
Price of inputs, USD (log(t-1))	-0.036 (0.022)	-0.089 (0.048)	-0.033 (0.053)	-0.016 (0.017)
Price of beef per kg, USD (log)	-0.022 (0.022)	0.001 (0.078)	-0.061 (0.047)	0.001 (0.020)
Price of beef per kg, USD (log(t-1))	-0.031 (0.015)	-0.113 (0.067)	0.033 (0.056)	-0.012 (0.015)
Labour employed in agriculture, people (log)	-0.049 (0.023)	-0.158 (0.065)	-0.021 (0.048)	-0.020 (0.019)
Number of agricultural machineries, units (log)	-0.007 (0.005)	0.006 (0.017)	0.003 (0.011)	-0.005 (0.005)
Number of cooperatives, units (log)	-0.000 (0.013)	0.022 (0.023)	0.006 (0.027)	0.004 (0.009)
Area of agricultural land, ha (log)	-0.012 (0.016)	-0.036 (0.026)	0.041 (0.024)	0.001 (0.014)
Constant	-0.530 (0.249)	-2.008 (0.622)	-0.612 (0.464)	-0.251 (0.213)
Observations	614	614	610	614
Number of id	84	84	84	84
First-order autocorrelation	0.00386	6.48e-05	1.68e-05	0.0002
Second-order autocorrelation	0.874	0.127	0.463	0.514
Hansen overID test	0.850	0.745	0.963	0.797

Note: Standard errors in parentheses.

TABLE A4 Lag selection for anti-corruption efforts based on least absolute shrinkage and selection operator (LASSO) estimation results for total cattle production growth.

	aic	aicc	bic	ebic
Anticorruption efforts				
-.	x	x		
L1.	x	x	x	x
L2.	x	x		
L4.	x	x		
L5.	x	x		
L6.	x	x		

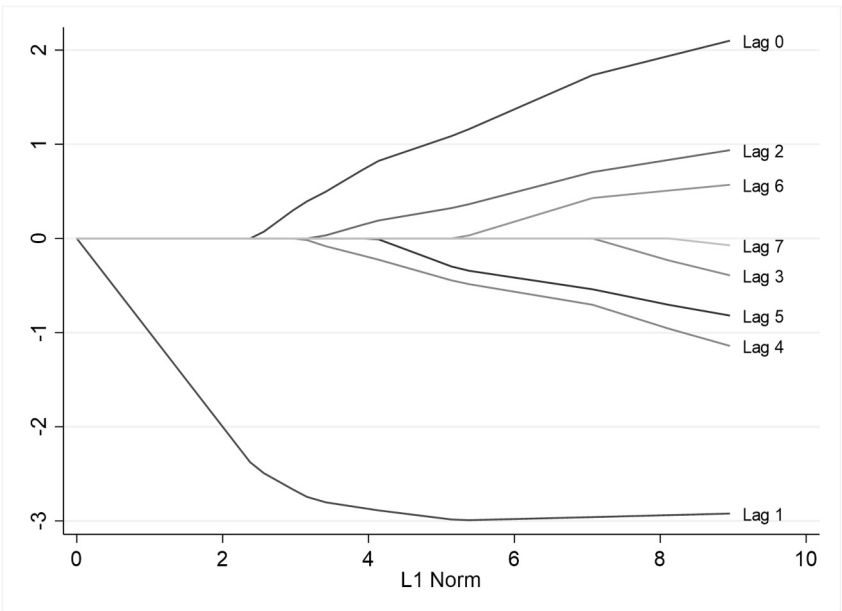


FIGURE A1 The coefficient path of each lagged value of anti-corruption efforts on total cattle production growth for the lasso penalty parameters.

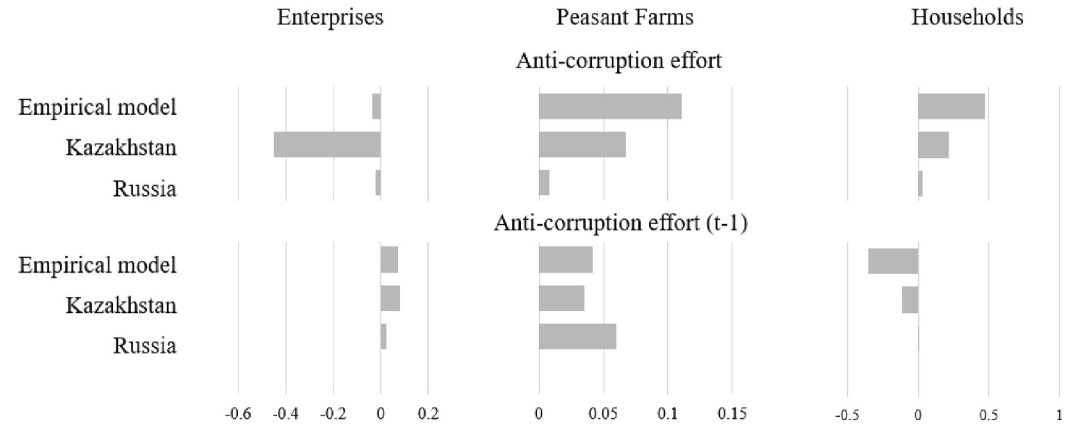


FIGURE A2 Estimated marginal effects for growth rates and anti-corruption efforts by the country.

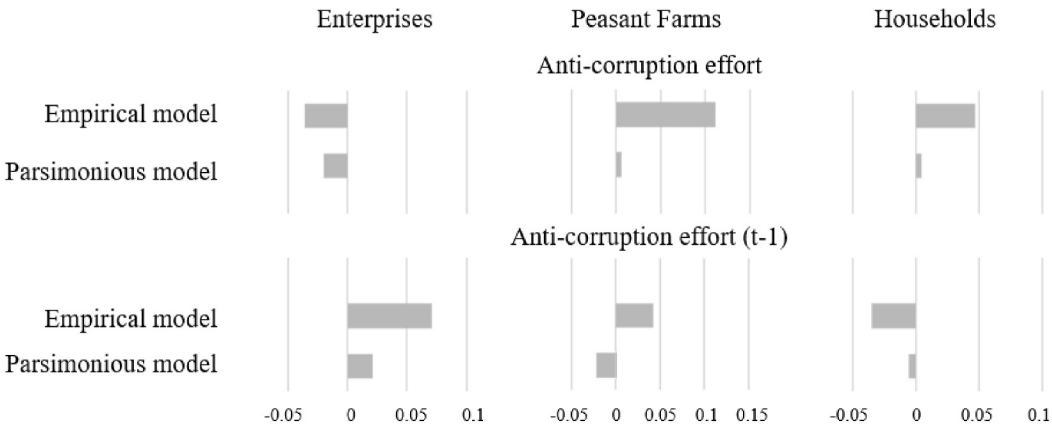


FIGURE A3 Estimated marginal effects for anticorruption efforts across farm types using the fixed effects model.